Separator Design

A Technical Design & Verification Report

Submitted by

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1. Introduction to Separators

A separator is a pressure vessel used in oil and gas production facilities to separate the well stream into gas and liquid phases (oil and water) based on differences in density and gravity. Its primary function is to remove free gas from liquids and free liquids from gas, producing stable crude oil and clean gas for further processing. Separators are sometimes called “gas scrubbers” when the gas-to-liquid ratio is very high, and “traps” when they handle flow directly from wells. All have the same configuration and are sized by the same procedure.

Objectives of separation

- Remove gas from crude oil.
- Remove liquid droplets from gas.
- Reduce downstream equipment load.
- Improve product quality.
- Protect compressors, pumps, and pipelines.

Factors affecting separation

The following must be determined before separator design:

- Gas and liquid flow rates (minimum, average, peak).
- Operating and design pressures and temperatures.
- Surging or slugging tendencies of the feed.
- Physical properties of the fluids (density, compressibility).
- Designed degree of separation (e.g. removing 100% of particles > 10 micron).
- Presence of impurities (paraffin, sand, scale).
- Foaming tendencies of the crude.
- Corrosive tendencies of the liquids or gas.

2. Equipment Description

Horizontal separators

Fluid enters the separator and hits an inlet diverter, causing a sudden change in momentum that gives the initial gross separation of liquid and vapour. Gravity causes the liquid droplets to fall to the bottom, where the liquid-collection section provides the retention time needed for entrained gas to evolve, plus surge volume for liquid slugs. Liquid leaves through a level-controlled dump valve. Gas flows horizontally through the gravity-settling section above the liquid; small drops not separated there are removed by a mist extractor (vanes, wire mesh or plates) before the gas leaves. A pressure controller maintains vessel pressure. Horizontal units are normally run half full of liquid to maximise the gas–liquid interface area.

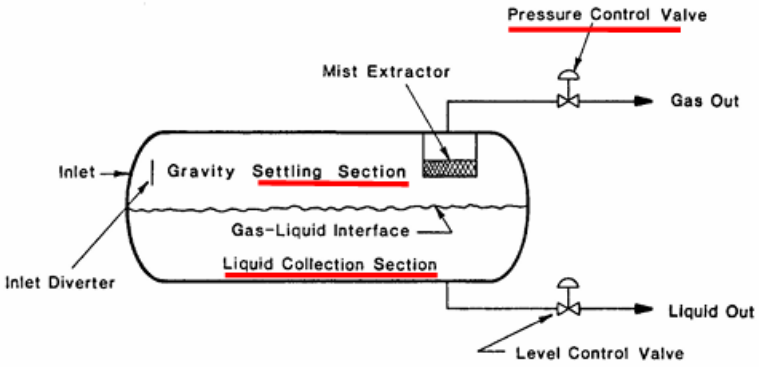


Figure 4-1. Horizontal separator schematic.

Vertical separators

Inlet flow enters through the side; the inlet diverter does the gross separation. Liquid flows down to the collection section and out through the level-controlled dump valve, while gas bubbles rise counter-current into the vapour space. Gas flows vertically upward through the gravity-settling section — where drops fall counter to the gas — and through the mist extractor before leaving.

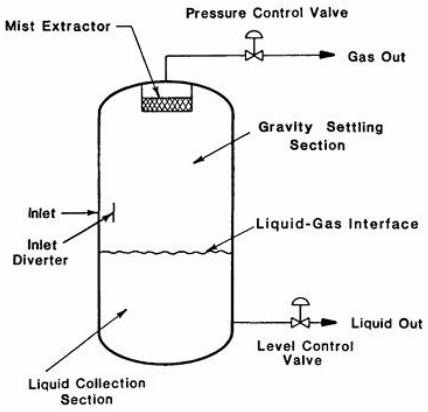


Figure 4-2. Vertical separator schematic.

Horizontal vs. vertical selection

Horizontal separators are smaller and less expensive for a given gas capacity: droplets fall perpendicular to the gas flow, the interface area is larger, and liquid capacity is greater. They are best for liquid–liquid separation and foaming crudes. Their drawbacks:

- Poorer solids handling (several drains needed along the length).
- More plan area required (important offshore).
- Less liquid surge margin; internal waves can trip high-level shut-downs.

Vertical vessels handle solids better (centre bottom drain) and need less plan area, but the relief valve and controls can be harder to service, and height may force removal from a skid for transport. Overall, horizontal vessels are most economical for normal oil–gas service (emulsions, foam, high GOR); vertical vessels suit low-GOR service and high-GOR scrubbers.

3. Theory of Settling

In the gravity-settling section a liquid drop settles at the velocity that balances the gravity (buoyant) force against the drag force of its motion through the gas. The drag force is $F_D = C_D \cdot A \cdot (\rho V_t^2/2)$, where C_D = drag coefficient, A = droplet cross-section, ρ = gas density, V_t = terminal velocity. For laminar (Stokes) flow $C_D = 24/Re$, but production-facility design needs the

full relation:

Eq. 1: $C_D = 24/Re + 3/\sqrt{Re} + 0.34$

Equating drag and buoyancy gives the terminal settling velocity:

Eq. 2: $V_t = 0.0119\ [\ ((\rho_L - \rho_g)/\rho_g)(d_m/C_D)\ ]^{0.5}$

where ρ_L, ρ_g = liquid and gas densities (lb/ft³), d_m = drop diameter (micron), μ = gas viscosity (cP). Equations 1 and 2 are solved iteratively:

1. Start with Eq. 2 (assume $C_D = 0.34$).
2. Calculate $Re = 0.0049\ \rho_g\ d_m\ V_t/\mu$.
3. From Re calculate C_D using Eq. 1.
4. Recalculate V_t using Eq. 2.
5. Return to step 2 and iterate until converged.

Design considerations

Drop size

The gas-separation section conditions the gas for the mist extractor. Removing 100-micron drops here keeps the mist extractor (which handles 10–100 micron) from flooding, so the gas capacity equations are based on 100-micron removal. Scrubbers may use 500 micron; vent and flare systems are sized for 300–500 micron in the gravity section.

Retention time

To let gas and liquid reach equilibrium at separator pressure, a certain liquid storage (retention time = liquid volume / liquid flow rate) is required. 30 s to 3 min is typical; up to four times that for foaming crude.

Re-entrainment

High gas velocity at the gas–liquid interface transfers momentum to the liquid, creating waves that shed droplets. The rule of thumb is to limit the slenderness ratio to 4–5 for half-full horizontal separators, especially above 1,000 psig or below 30 °API.

4. Two-Phase Separator Sizing

Horizontal separators (50% full)

Eq. 3 (gas): $d \cdot L_{eff} = 420\ (T \cdot Z \cdot Q_g/P)[(\rho_g/(\rho_L - \rho_g))(C_D/d_m)]^{0.5}$

Eq. 4 (liquid): $d^2 \cdot L_{eff} = 1.42\ (t_r \cdot Q_l)$

Eq. 5 (seam-to-seam): $L_{ss} = L_{eff} + d/12$ *(gas)* ; $L_{ss} = (4/3)L_{eff}$ *(liquid)*

d = diameter (in), L_{eff} = effective length (ft), T = °R, Q_g = MMscfd, P = psia, Z = compressibility, d_m = drop size (micron). The interface must stay ≥ 10 in from the bottom and 16 in from the top ⇒ minimum diameter 26 in.

Procedure — horizontal two-phase

1. Calculate d, L_{eff} combinations satisfying gas capacity (Eq. 3).
2. Calculate d, L_{eff} combinations satisfying retention time (Eq. 4).
3. Estimate seam-to-seam length (Eq. 5).
4. Select a reasonable diameter and length; slenderness ratio ($12\ L_{ss}/d$) of 3–4 is common, not exceeding 5 without further re-entrainment study.

Vertical separators

Eq. 6 (gas): $d^2 = 5040\ (T \cdot Z \cdot Q_g/P)[(\rho_g/(\rho_L - \rho_g))(C_D/d_m)]^{0.5}$

Eq. 7 (liquid): $d^2 \cdot h = t_r \cdot Q_l / 0.12$

Eq. 8 (seam-to-seam): L_{ss} = larger of (h+76)/12 or (h+d+40)/12

The constant 5040 is fixed by this unit set and is not a free parameter. The interface should be ≥ 2 ft from the bottom and ≥ 4 ft from the top ⇒ minimum height 6 ft.

Procedure — vertical two-phase

1. Calculate the minimum diameter from gas capacity (Eq. 6).
2. Compute d and liquid height h combinations for various retention times (Eq. 7).
3. Compute seam-to-seam length (Eq. 8).
4. Compute slenderness ratio; choose a reasonable size with d greater than the gas-capacity minimum (SR 3–4).

5. Worked Examples — Two-Phase Separators

Common data: 10 MMscfd at 0.6 gas SG, 2,000 bopd at 40 °API, P = 1,000 psia, T = 60 °F (520 °R), Z = 0.84, ρ_L = 51.5 lb/ft³, d_m = 140 micron, μ = 0.013 cP.

5.1 Vertical two-phase

Step 1 — Gas density: $\rho_g = 2.7 \cdot \gamma_g \cdot P/(T \cdot Z) = 2.7 \times 0.6 \times 1000/(520 \times 0.84) = 3.71$ lb/ft³

Step 2 — Drag coefficient (iterate Eqs. 1–2):

Iteration	C_D	V_t (ft/s)	Re
1 (assumed)	0.340	0.867	170
2	0.712	0.599	117
3	0.822	0.558	109
converged	0.854	0.547	107

Step 3 — Gas capacity (Eq. 6): $d^2 = 5040(520 \times 0.84 \times 10/1000)[(3.71/47.79)(0.854/140)]^{0.5} = 478 \Rightarrow d_{min} = 21.8$ in

Step 4 — Liquid capacity (Eq. 7), t_r = 3 min: $d^2 \cdot h = 3 \times 2000/0.12 = 50{,}000$

Step 5 — Combinations of d, h and seam-to-seam length (Eq. 8):

d (in)	h (in)	L_ss (ft)	12·L_ss/d
30	55.6	11.0	4.4
36	38.6	9.6	3.2
42	28.3	9.2	2.6
48	21.7	9.1	2.3

Table 4-1 — Vertical two-phase example, diameter vs. height.

Step 6 — Selection: a 36 in × 10 ft vertical separator gives slightly more than 3 min retention, with d > 21.8 in and a slenderness ratio of 3.2.

5.2 Horizontal two-phase

Using the same fluids, C_D = 0.854.

Step 1 — Gas capacity (Eq. 3): $d \cdot L_{eff} = 420(520 \times 0.84 \times 10/1000)[(3.71/47.79)(0.854/140)]^{0.5} = 39.9$

Step 2 — Liquid capacity (Eq. 4), t_r = 3 min: $d^2 \cdot L_{eff} = 1.42 \times 3 \times 2000 = 8{,}520$

Step 3 — Seam-to-seam length (Eq. 5) and slenderness ratio for each diameter:

d (in)	Gas L_eff (ft)	Liquid L_eff (ft)	L_ss (ft)	12·L_ss/d
16	2.5	33.5	44.7	33.5
20	2.0	21.4	28.5	17.1
24	1.7	14.9	19.9	9.9
30	1.3	9.5	12.7	5.1
36	1.1	6.6	9.1*	3.0
42	0.9	4.9	7.4*	2.1
48	0.8	3.7	6.2*	1.6

Table 4-2 — Horizontal two-phase example (L_ss = L_eff + 2.5 governs).*

Step 4 — Selection: a 36 in × 10 ft horizontal separator provides about 3 min retention at a slenderness ratio of 3.0.

6. Three-Phase Separator Sizing

Three-phase separators split gas, oil and water. The oil pad is sized so that 500-micron water droplets settle out (giving < 5–10% water in the oil); separating oil from water is easier and is not the governing criterion. With no lab data, a 10-minute retention time is used for both oil and water at maximum flow.

Horizontal three-phase

Oil pad: $(h_o)_{max} = 0.00128\,(t_r)_o(\Delta SG)d_m^2/\mu_o$

Gas: $d\cdot L_{eff} = 420\,(T\cdot Z\cdot Q_g/P)[(\rho_g/(\rho_L-\rho_g))(C_D/d_m)]^{0.5}$

Liquid: $d^2\cdot L_{eff} = 1.42[Q_w(t_r)_w + Q_o(t_r)_o]$

Procedure — horizontal three-phase

1. Select $(t_r)_o$ and $(t_r)_w$.
2. Calculate $(h_o)_{max}$ (use 500-micron droplet).
3. Calculate $A_w/A = 0.5\,Q_w(t_r)_w/[(t_r)_oQ_o+(t_r)_wQ_w]$.
4. Determine h_o/d from the design curve.
5. Calculate $d_{max} = (h_o)_{max}/(h_o/d)$.
6. Calculate d, L_{eff} combinations (< d_{max}) for gas capacity (100-micron).
7. Compute d, L_{eff} combinations for liquid retention.
8. Compute seam-to-seam length and slenderness ratio (3–5).
9. Graph the results and choose a size that violates neither the gas-capacity nor the oil-pad limit.

Vertical three-phase

Water settling (min. diameter): $d^2 = 6690\,(Q_o\cdot\mu_o)/(\Delta SG\cdot d_m^2)$

Retention (height): $h_o+h_w = [(t_r)_oQ_o + (t_r)_wQ_w]/(0.12\,d^2)$

Procedure — vertical three-phase

1. Calculate the difference in specific gravities, ΔSG.
2. Calculate the minimum diameter for water-droplet settling.
3. Apply the liquid-retention constraint.
4. Compute d and (h_o+h_w) combinations for diameters greater than d_{min}.

5. Compute seam-to-seam length (larger of the two geometry estimates).
6. Compute slenderness ratio (1.5–3 common).
7. Repeat for various retention times if necessary, and graph.
8. Choose a reasonable size.

7. Worked Examples — Three-Phase Separators

Common data: 5 MMscfd, 5,000 bopd at 30 °API (SG_o = 0.876), 3,000 bwpd (SG_w = 1.07), P = 100 psia, T = 90 °F, μ_o = 10 cP, d_m = 500 micron, retention 10 min each.

7.1 Vertical three-phase

Step 1 — Difference in SG: $\Delta SG = 1.07 - 0.876 = 0.194$

Step 2 — Minimum diameter for water settling: $d^2 = 6690(5000\times10)/(0.194\times500^2) = 6,897 \Rightarrow d_{min} = 83.0$ in

Step 3 — Retention height: $h_o+h_w = (10\times5000+10\times3000)/(0.12\ d^2) = 666,667/d^2$

d (in)	h_o + h_w (in)	(ft)
84	94.5	7.9
90	82.3	6.9
96	72.3	6.0

Table 5-1 — Vertical three-phase, height vs. diameter (d ≥ d_min).

Step 4 — Selection: a 90 in × 15 ft (or 96 in × 12 ft 6 in) vertical three-phase separator, slenderness ratio 1.5–3.

7.2 Horizontal three-phase

Step 1 — Oil pad: $(h_o)_{max} = 0.00128\times10\times0.194\times500^2/10 = 62$ in; the resulting d_{max} is far above any practical diameter, so it is not limiting.

Step 2 — Gas capacity (Eq. 3): $d\cdot L_{eff} = 102 \Rightarrow L_{eff}$ of only 1–2 ft (gas never governs).

Step 3 — Liquid retention: $d^2\cdot L_{eff} = 1.42[(3000)(10)+(5000)(10)] = 113,600$

d (in)	L_eff (ft)	L_ss (ft)	12·L_ss/d
60	31.6	42.1	8.4
72	21.9	29.2	4.9
84	16.1	21.5	3.1
96	12.3	16.4	2.1
108	9.7	13.0	1.4

Table 5-3 — Horizontal three-phase example (t_r = 10 min).

Step 4 — Selection: a 90 in × 20 ft (or 96 in × 17 ft 6 in, or 102 in × 15 ft) horizontal three-phase separator at a slenderness ratio of 3–5.

8. Field Verification — Vertical Separator (Gas-Capacity Governed)

Gas flow rate	179,000 kg/hr
Liquid flow rate	7,200 kg/hr
Operating pressure	24 kg/cm² (g)
Operating temperature	33 °C
Gas density (operating)	16 kg/m³
Liquid density	750 kg/m³

Design droplet size	90 micron
Gas viscosity	0.0119 cP
Z	0.84

Step 1 — Convert to field units

$\rho_g = 16/16.018 = 0.999$ lb/ft³ ; $\rho_L = 750/16.018 = 46.82$ lb/ft³

$P = 24\times14.223 + 14.7 = 356$ psia ; $T = (33\times9/5+32)+460 = 551.4$ °R

Actual gas = 179,000/16 = 11,188 m³/hr = 109.7 ft³/s ; standard gas ≈ 216.5 MMscfd

Step 2 — Drag coefficient (iterate Eqs. 1–2)

Iteration	C_D	V_t (ft/s)	Re
1 (assumed)	0.340	1.311	48.6
2	1.265	0.680	25.2
3	1.892	0.556	20.6
4	2.167	0.519	19.2
5	2.272	0.507	18.8
converged	2.331	0.501	18.5

Step 3 — Gas-capacity diameter

Design basis (mist extractor, Souders–Brown): $V_{max} = K\sqrt{[(\rho_L-\rho_g)/\rho_g]} = 0.283\times\sqrt{(45.82/0.999)} = 1.92$ ft/s

$A = 109.7/1.92 = 57.3$ ft² ; $d = \sqrt{(4A/\pi)} = 102.5$ in (2,604 mm)

Cross-check by 90-micron gravity settling (Eq. 6, correct constant 5040, $C_D = 2.33$) gives d ≈ 184 in. That value assumes no mist extractor and is over-conservative; the mist-extractor diameter (102.5 in) is the design value. Using a non-standard constant (e.g. 1200) to force d = 87.5 in is not dimensionally valid. K = 0.283 is within the GPSA range 0.18–0.35.

Step 4 — Liquid capacity

$Q_l = 7{,}200/750 = 9.6$ m³/hr = 1,449 BPD ; $d^2{\cdot}h = t_r Q_l/0.12$ (small height for d = 102.5 in).

Step 5 — Verification against the existing vessel

Existing diameter	2,600 mm = 102.4 in
Required diameter (Souders–Brown)	102.5 in
Existing height	3,600 mm = 11.8 ft
Minimum height required	~11 ft

Result: the existing 2,600 × 3,600 mm separator sits essentially at the gas-capacity design point (102.4 vs 102.5 in) and exceeds the minimum height — it is correctly sized.

9. Field Verification — Horizontal Three-Phase Separator

Gas flow rate	35 MMSCFD
Oil flow rate	1,250 BOPD
Water flow rate	750 BWPD
Operating pressure	12 kg/cm² (g)
Operating temperature	25 °C
Gas density	9.2 kg/m³
Oil density	833 kg/m³
Water density	1,060 kg/m³
Oil viscosity	4.72 cSt

Design droplet size	500 micron
Oil & water retention	10 min each

Step 1 — Fluid properties

$SG_o = 0.833$; $\mu_o = 4.72 \times 0.833 = 3.93$ cP ; $SG_w = 1.060$; $\Delta SG = 0.227$

Step 2 — Oil-pad-thickness check

$(h_o)_{max} = 0.00128 \times 10 \times 0.227 \times 500^2 / 3.93 = 184.9$ in ; $D_{max} = 184.9/0.257 = 719$ in — not limiting

Step 3 — Gas capacity

$\rho_g = 0.574$, $\rho_L = 52.0$ lb/ft³, $P = 185.4$ psia, $T = 537$ °R ; $d \cdot L_{eff} = 135 \rightarrow L_{eff}$ of 1.4–2.8 ft (gas never governs)

Step 4 — Liquid retention (governs)

$d^2 \cdot L_{eff} = 1.42[(750)(10)+(1250)(10)] = 28{,}400$

d (in)	L_eff (ft)	L_ss (ft)	SR = 12·L_ss/d
48	12.33	16.44	4.11
60	7.89	10.52	2.10
72	5.48	7.31	1.22

The bare-minimum 72 in × 7.3 ft satisfies retention but has SR = 1.2 (too low); a horizontal vessel needs SR = 3–5, so a longer vessel is selected.

Step 5 — Verification against the existing vessel

Existing separator	72 in × 20 ft (1,829 × 6,100 mm)
Existing slenderness ratio	12×20/72 = 3.3 (within 3–5)
Required effective length at 72 in	5.5 ft (provided 15 ft)
Gas capacity	satisfied (never governs)

Result: the existing 72 in × 20 ft separator provides far more retention length than required (15 ft vs 5.5 ft) at a proper slenderness ratio of 3.3 — it comfortably satisfies both constraints.

10. Summary of Field Verification

Item	Vertical (field)	Horizontal (field)
Governing constraint	Gas capacity	Liquid retention
Required diameter	102.5 in	72 in (SR 3–5)
Required length / height	~11 ft	≥ 5.5 ft eff (use ~20 ft)
Existing vessel	2,600 × 3,600 mm	72 in × 20 ft
Verdict	Adequate	Adequate

Both existing separators are confirmed adequate for the specified duties using the standard Arnold & Stewart sizing methods, with the mist-extractor (Souders–Brown) basis used for the vertical gas-capacity diameter.

Method basis: Arnold & Stewart, Surface Production Operations, Vol. 1; Souders–Brown (1934) / GPSA Engineering Data Book.